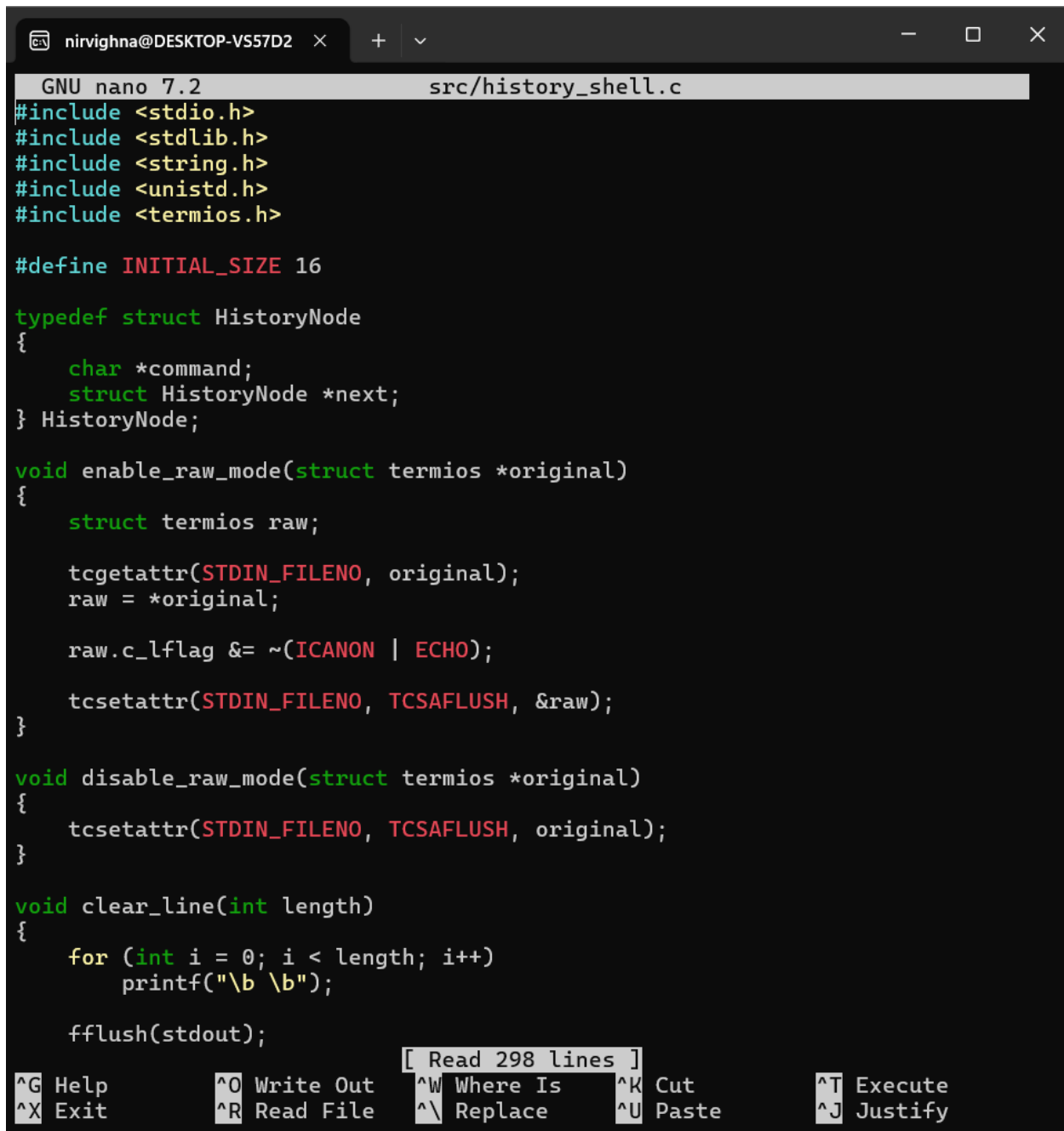


SKILL 11:



```
GNU nano 7.2 src/history_shell.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <termios.h>

#define INITIAL_SIZE 16

typedef struct HistoryNode
{
    char *command;
    struct HistoryNode *next;
} HistoryNode;

void enable_raw_mode(struct termios *original)
{
    struct termios raw;

    tcgetattr(STDIN_FILENO, original);
    raw = *original;

    raw.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
}

void disable_raw_mode(struct termios *original)
{
    tcsetattr(STDIN_FILENO, TCSAFLUSH, original);
}

void clear_line(int length)
{
    for (int i = 0; i < length; i++)
        printf("\b \b");

    fflush(stdout);
}
```

[Read 298 lines]

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^\\ Replace	^U Paste	^J Justify

```
nirvighna@DESKTOP-VS57D2  x  +  v  -  □  ×  
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ nano src/history_shell.c  
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ #include <stdio.h>  
#include <string.h>  
  
#define MAX_HISTORY 5  
#define MAX_LENGTH 100  
  
char history[MAX_HISTORY][MAX_LENGTH];  
int count = 0;  
  
void add_history(char *command)  
{  
    if (count < MAX_HISTORY)  
    {  
        strcpy(history[count], command);  
        count++;  
    }  
    else  
    {  
        for (int i = 0; i < MAX_HISTORY - 1; i++)  
            strcpy(history[i], history[i + 1]);  
  
        strcpy(history[MAX_HISTORY - 1], command);  
    }  
}  
  
void show_history()  
{  
    printf("\nCommand History:\n");  
  
    for (int i = 0; i < count; i++)  
        printf("%d  %s\n", i + 1, history[i]);  
}  
  
int main()  
{  
    char command[MAX_LENGTH];  
  
    printf("History Shell\n");  
    printf("Type commands. Type 'history' to display history.\n");  
    printf("Type 'exit' to quit.\n\n");
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./bin/history_shell
Interactive History Shell
Type 'exit' to quit.

history-shell> ls
You entered: ls
history-shell> pwd
You entered: pwd
history-shell> date
You entered: date
history-shell> whoami
You entered: whoami
history-shell> history
You entered: history
history-shell> exit
Shell terminated.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```

TASK :2



```
GNU nano 7.2 src/pipeline_shell.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];
    pid_t p1, p2;

    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        return 1;
    }

    p1 = fork();

    if (p1 == -1)
    {
        perror("fork");
        return 1;
    }

    if (p1 == 0)
    {
        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[0]);
        close(pipefd[1]);

        execlp("ls", "ls", NULL);

        perror("execlp");
        exit(1);
    }
}
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ gcc -Wall -Wextra -g src/pipeline_shell.c -o bin/pipeline_shell
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./bin/pipeline_shell
14
Pipeline completed successfully.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l bin/history_shell bin/pipeline_shell
-rwxr-xr-x 1 nirvighna nirvighna 22088 Sep 28 04:01 bin/history_shell
-rwxr-xr-x 1 nirvighna nirvighna 18240 Sep 28 04:04 bin/pipeline_shell
```

SKILL 12:



```
GNU nano 7.2 pipeline1.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];
    pid_t pid1, pid2;

    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        exit(1);
    }

    pid1 = fork();

    if (pid1 == -1)
    {
        perror("fork");
        exit(1);
    }

    if (pid1 == 0)
    {
        // First child: ls

        close(pipefd[0]);

        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[1]);

        execlp("ls", "ls", NULL);

        perror("exec ls");
    }
}
```

Terminal Output:

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./pipeline1
16
Pipeline completed successfully.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$
```

nirvighna@DESKTOP-VS57D2

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nirvighna@DESKTOP-VS57D2I:~/OS-Week1\$ cat pipeline1.c

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];
    pid_t pid1, pid2;

    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        exit(1);
    }

    pid1 = fork();

    if (pid1 == -1)
    {
        perror("fork");
        exit(1);
    }

    if (pid1 == 0)
    {
        // First child: ls

        close(pipefd[0]);

        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[1]);

        execlp("ls", "ls", NULL);

        perror("exec ls");
        exit(1);
    }
}
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./bin/pipeline_shell
14
Pipeline completed successfully.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ nano pipeline1.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ gcc pipeline1.c -o pipeline1
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./pipeline1
16
Pipeline completed successfully.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls | wc -l
16
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls | wc -l
16
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ touch test1.txt test2.txt test3.txt
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./pipeline1
19
Pipeline completed successfully.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls | wc -l
19
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ cat pipeline1.c
```

```
nirvighna@DESKTOP-VS57D2  ×  +  ∨
GNU nano 7.2                pipeline2.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pipe1[2];
    int pipe2[2];

    pid_t pid1, pid2, pid3;

    // Create first pipe
    if (pipe(pipe1) == -1)
    {
        perror("pipe1");
        exit(1);
    }

    // Create second pipe
    if (pipe(pipe2) == -1)
    {
        perror("pipe2");
        exit(1);
    }

    // First child: ls
    pid1 = fork();

    if (pid1 == -1)
    {
        perror("fork");
        exit(1);
    }

    if (pid1 == 0)
    {
        ^G Help      ^O Write Out    ^W Where Is    ^K Cut         ^T Execute
        ^X Exit      ^R Read File    ^\ Replace     ^U Paste       ^J Justify
    }
}
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ nano pipeline2.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ gcc pipeline2.c -o pipeline2
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./pipeline2
9
Long pipeline completed successfully.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls | grep .c | wc -l
9
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ touch a.c b.c c.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./pipeline2
12
Long pipeline completed successfully.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls
Makefile  mini_shell.c  pipeline2.c  test2.txt
a.c       nano.2750.save producer_consumer  test3.txt
b.c       output.txt    producer_consumer.c  wait_demo
bin       pipeline1     src              wait_demo.c
c.c       pipeline1.c   strace_output.txt   zombie
input.txt pipeline2     test1.txt          zombie.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```